

Todd Takala

Lead Data Scientist — Lead Machine Learning Engineer

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Summary of Qualifications

Lead Data Scientist with a proven track record of delivering over \$1.3 billion in business value by leveraging machine learning and end-to-end data science solutions to solve complex challenges. Expertise in managing the complete project lifecycle, from stakeholder engagement and KPI definition to development and deployment.

- **Machine Learning & AI Leadership:** Spearheaded the creation of an ML/AI system to predict equipment repair times, resulting in an \$80 million reduction in warranty costs and a CEO Award. Designed and optimized winning regression and recommendation systems, achieving two-time champion status in the Amazon Machine Learning University competition.
- **MLOps & End-to-End Pipelines:** Engineered and deployed robust ETL/MLOps pipelines using Snowflake and AWS services (SageMaker, S3, Step Functions) to automate model retraining. Championed MLOps best practices, setting new standards for GitHub CI/CD, Docker, and AWS architecture. Developed an end-to-end MLOps pipeline to deploy AI solutions for analyzing IoT and telematics data.
- **Data Analysis & Engineering:** Resolved critical design flaws in mining equipment through reproducible analysis in R and SQL, saving clients \$1.2 billion over six years. Created a regression model that identified key performance factors, leading to \$62 million in savings within 90 days. Analyzed over 2 petabytes of IoT data from a 400,000-robot fleet to establish a new design-for-reliability benchmark.
- **Team Leadership & Technical Mentorship:** Directed a high-performance team of up to 8 engineers, fostering a data-driven approach to solve equipment performance issues. Provided technical leadership by developing foundational predictive forecasting models for supply chain management.

Professional Experience

Lead Data Scientist — Caterpillar Inc. — Surprise, AZ September 2022 – Present

Led the end-to-end development of high-impact data science initiatives, including an award-winning ML system to predict equipment repair times. Engineered and deployed robust MLOps pipelines using AWS and Snowflake and established best practices for CI/CD and system architecture. Managed project lifecycles and created comprehensive monitoring plans to validate business impact.

Reliability Analytics Manager — Amazon — Tempe, AZ February 2022 – September 2022

Pioneered a new design-for-reliability benchmark by analyzing petabyte-scale IoT data from a large robotics fleet. Authored an influential white paper on complex Weibull analysis and secured adoption of the new benchmark by senior leadership.

Data Engineering / Reliability Manager — Empire-Cat — Mesa, AZ January 2021 – February 2022

Led a team to establish an MLOps pipeline with ARIMA and SVM for AI-driven forecasting in supply chain management, integrated data, managed a component exchange program, collaborated with clients on durability forecasting, and addressed design flaws in mining equipment, reducing client costs by \$1.2B. This involved creating a comprehensive machine learning dataset from NLP on work orders and IoT telemetry.

Core Skills

Python, Redshift, ECR, DBA, ETL, TensorFlow, PyTorch, Scikit-learn, Transformers, Anomaly Detection, Experiments, pandas, keras, numpy, Git, Strategic Thinking, Adaptability, Agile, Continuous Integration, Programming Languages, Distributed Computing, Architectural Design, Statistical Tools, Data Gathering, Willing to Learn

Education

Bachelor of Science in Mechanical Engineering — Arizona State University — Tempe, AZ 2006
STEM coursework included: Numerical Methods, Computational Thermodynamics, Statistics, Logic, Linear Algebra, Finite Element Analysis

Certifications

Data Camp: Data Scientist Professional Certificate, Certified SQL Associate, AI Fundamentals Certificate

Canonical: Ubuntu Linux Professional Certificate

GitHub: Career Essentials in GitHub Professional Certificate

Udemy: Streamlit for Snowflake Masterclass

BerkeleyX: The Science of Happiness at Work Professional Certificate

Empire-Cat: Lean 6 Sigma Green Belt

Komatsu North America: Certified Technical Communicator

Society of Automotive Engineers: Cost and Finance for Engineers

Arizona State Board of Technical Registration: EIT, Mechanical Engineering – Registration No. 10670